温州大学数理与电子信息工程学院

路由与交换 实验报告

备注：实验包括基本技能、进阶技能和创新技能，如果没有创新技能方案则最高分为 90 分，

实验名称:	实验七　BGP 的配置与管理				
班　　级:		姓　　名:	刘军	学　　号:	
实验地点:		日　　期:			

一、实验目的：

- 了解 BGP 协议的工作原理，理解 BGP 对等体的概念与类型
- 理解 BGP 的路径选择过程，掌握应用路由映射实现策略路由的方法
- 掌握 BGP 协议的规划、配置、测试与故障排除
-

二、基本技能实验内容、要求和环境：

如图所示的网络拓扑，要求采用 BGP 与 RIP 协议的组合使得该网络中的所有网段之间能够相互通信。

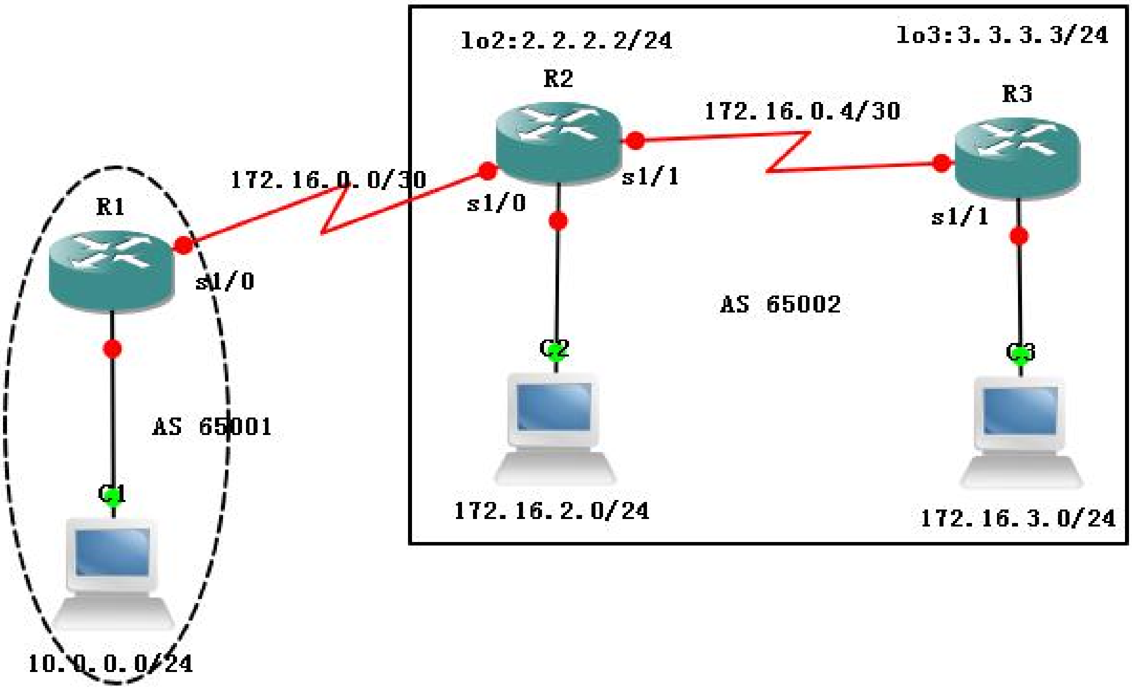

表 1　“BGP 的配置与管理”之相关规划

规划内容	规划要点	参考建议
网络环境的搭建	线缆的选择、网络设备型号、网络设备数以及网络设备之间的物理连接	略
路由器的基本配置	路由器主机名，特权密码、Telnet，路由器接口的 IP 地址	略
IP 地址的分配	每个网段的 IP 网络号，PC 的 IP 地址、子网掩码、默认网关	见图 6.9.1
RIP 的配置	参与 RIP 更新的路由器与接口	规划提示：仅自治系统 65002 中的路由器 R2 与 R3 启动 RIP 路由进程 路由器 R2 参与更新的网络有 172.16.0.4/30、172.16.2.0/24 与 2.2.2.0/24

		路由器 R3 参与更新的网络有 172.16.0.4/30、172.16.3.0/24 与 3.3.3.0/24
BGP 的配置	自治系统划分、启动 BGP 的路由器、邻居与需要宣告的网络	规划提示：所有路由器启动 BGP 路由进程，其中 R1 属于自治系统 65001，R2 与 R3 属于自治系统 65002 路由器 R1 上被宣告的直连路由有：10.0.0.0/24 路由器 R2 上被宣告的直连路由有：172.16.2.0/24 路由器 R2 上被宣告的直连路由有：172.16.3.0/24
路由的测试	通过主机测试 通过路由器测试	ping、tracert ping、扩展 ping、traceroute show ip bgp summary、show ip bgp neighbors

三、基本技能实验结果与分析：

1. 配置路由器名称与接口 IP 地址：

R1#show ip interface brief

Interface	IP-Address	OK? Method Status	Protocol
FastEthernet0/0	**10.0.0.1**	**YES manual up**	**up**
FastEthernet0/1	unassigned	YES unset administratively down	down
Serial1/0	**172.16.0.1**	**YES manual up**	**up**
Serial1/1	unassigned	YES unset administratively down	down
Serial1/2	unassigned	YES unset administratively down	down
Serial1/3	unassigned	YES unset administratively down	down

R2#show ip interface brief

Interface	IP-Address	OK? Method Status	Protocol
FastEthernet0/0	**172.16.2.1**	**YES manual up**	**up**
FastEthernet0/1	unassigned	YES unset administratively down	down
Serial1/0	**172.16.0.2**	**YES manual up**	**up**
Serial1/1	**172.16.0.5**	**YES manual up**	**up**
Serial1/2	unassigned	YES unset administratively down	down
Serial1/3	unassigned	YES unset administratively down	down
Loopback2	**2.2.2.2**	**YES manual up**	**up**

R3#show ip interface brief

Interface	IP-Address	OK? Method Status	Protocol
FastEthernet0/0	**172.16.3.1**	**YES manual up**	**up**
FastEthernet0/1	unassigned	YES unset administratively down	down
Serial1/0	unassigned	YES unset administratively down	down
Serial1/1	**172.16.0.6**	**YES manual up**	**up**
Serial1/2	unassigned	YES unset administratively down	down
Serial1/3	unassigned	YES unset administratively down	down
Loopback3	**3.3.3.3**	**YES manual up**	**up**

2. 配置主机 IP 地址、掩码与网关

```
VPCS[1]> ip 10.0.0.254/24 10.0.0.1
Checking for duplicate address...
PC1 : 10.0.0.254 255.255.255.0 gateway 10.0.0.1

VPCS[1]> 2
VPCS[2]> ip 172.16.2.254/24 172.16.2.1
Checking for duplicate address...
PC2 : 172.16.2.254 255.255.255.0 gateway 172.16.2.1

VPCS[2]> 3
VPCS[3]> ip 172.16.3.254/24 172.16.3.1
Checking for duplicate address...
PC3 : 172.16.3.254 255.255.255.0 gateway 172.16.3.1
```

3. 检查连通性

主机与网关:

```
VPCS[3]> ping 172.16.3.1
172.16.3.1 icmp_seq=1 ttl=255 time=27.500 ms
172.16.3.1 icmp_seq=2 ttl=255 time=25.001 ms
172.16.3.1 icmp_seq=3 ttl=255 time=30.001 ms
172.16.3.1 icmp_seq=4 ttl=255 time=25.000 ms
172.16.3.1 icmp_seq=5 ttl=255 time=20.000 ms

VPCS[3]> 2
VPCS[2]> ping 172.16.2.1
172.16.2.1 icmp_seq=1 ttl=255 time=30.001 ms
172.16.2.1 icmp_seq=2 ttl=255 time=20.000 ms
172.16.2.1 icmp_seq=3 ttl=255 time=27.500 ms
172.16.2.1 icmp_seq=4 ttl=255 time=22.500 ms
172.16.2.1 icmp_seq=5 ttl=255 time=25.001 ms

VPCS[2]> 1
VPCS[1]> ping 10.0.0.1
10.0.0.1 icmp_seq=1 ttl=255 time=27.501 ms
10.0.0.1 icmp_seq=2 ttl=255 time=25.000 ms
10.0.0.1 icmp_seq=3 ttl=255 time=17.500 ms
10.0.0.1 icmp_seq=4 ttl=255 time=27.501 ms
10.0.0.1 icmp_seq=5 ttl=255 time=27.501 ms
```

相邻路由器之间:

R2#ping 172.16.0.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 172.16.0.1, timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/25/56 ms

R2#ping 172.16.0.6

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 172.16.0.6, timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 16/20/28 ms

4. 配置 EBGP 对等体

R1(config)#router bgp 65001

R1(config-router)#neighbor 172.16.0.2 remote-as 65002

R1(config-router)#network 10.0.0.0 mask 255.255.255.0

R1(config-router)#exit

R1(config)#

***Mar 1 00:25:45.531: %BGP-5-ADJCHANGE: neighbor 172.16.0.2 Up**

R2(config)#router bgp 65002
R2(config-router)#neighbor 172.16.0.1 remote-as 65001
R2(config-router)#
***Mar 1 00:25:44.195: %BGP-5-ADJCHANGE: neighbor 172.16.0.1 Up**
R2(config-router)#exit

5. 配置 IBGP 对等体

配置指向环回接口的静态路由：

R2(config)#ip route 3.3.3.0 255.255.255.0 s1/1

R3(config)#ip route 2.2.2.0 255.255.255.0 s1/1

使用环回接口配置 IBGP 对等体：

R2(config)#router bgp 65002
R2(config-router)#neighbor 3.3.3.3 remote-as 65002
R2(config-router)#neighbor 3.3.3.3 update-source lo2
R2(config-router)#neighbor 3.3.3.3 next-hop-self
R2(config-router)#network 172.16.2.0 mask 255.255.255.0
R2(config-router)#exit
R2(config)#
***Mar 1 00:36:16.895: %BGP-5-ADJCHANGE: neighbor 3.3.3.3 Up**

R3(config)#router bgp 65002
R3(config-router)#neighbor 2.2.2.2 remote-as 65002
R3(config-router)#
***Mar 1 00:36:15.539: %BGP-5-ADJCHANGE: neighbor 2.2.2.2 Up**
R3(config-router)#neighbor 2.2.2.2 update-source lo3
R3(config-router)#neighbor 2.2.2.2 next-hop-self
R3(config-router)#network 172.16.3.0 mask 255.255.255.0
R3(config-router)#exit
R3(config)#

6. 检查路由表以及 BGP 对等体关系

```
R1#show ip route | include B
     172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
B        172.16.2.0/24 [20/0] via 172.16.0.2, 00:04:27
B        172.16.3.0/24 [20/0] via 172.16.0.2, 00:02:36
R1#show ip bgp
   Network          Next Hop            Metric LocPrf Weight Path
*> 10.0.0.0/24      0.0.0.0                  0         32768 i
*> 172.16.2.0/24    172.16.0.2               0             0 65002 i
*> 172.16.3.0/24    172.16.0.2                             0 65002 i
R1#show ip bgp summary
BGP router identifier 172.16.0.1, local AS number 65001
Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.16.0.2      4 65002      19      18        4    0    0 00:14:59        2

R2#show ip route | include B
B        172.16.3.0/24 [200/0] via 3.3.3.3, 00:06:14
     10.0.0.0/24 is subnetted, 1 subnets
```

B 10.0.0.0 [20/0] via 172.16.0.1, 00:17:03

R2#show ip bgp

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.0.0.0/24	172.16.0.1	0		0	65001 i
*> 172.16.2.0/24	0.0.0.0	0		32768	i
*>i172.16.3.0/24	3.3.3.3	0	100	0	i

R2#show ip bgp summary

BGP router identifier 2.2.2.2, local AS number 65002

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
3.3.3.3	4	65002	12	13	4	0	0	00:07:09	1
172.16.0.1	4	65001	21	22	4	0	0	00:17:42	1

R3#show ip route | include B

B 172.16.2.0/24 [200/0] via 2.2.2.2, 00:09:29

10.0.0.0/24 is subnetted, 1 subnets

B 10.0.0.0 [200/0] via 2.2.2.2, 00:09:29

R3#show ip bgp

Network	Next Hop	Metric	LocPrf	Weight	Path
*>i10.0.0.0/24	2.2.2.2	0	100	0	65001 i
*>i172.16.2.0/24	2.2.2.2	0	100	0	i
*> 172.16.3.0/24	0.0.0.0	0		32768	i

R3#show ip bgp summary

BGP router identifier 3.3.3.3, local AS number 65002

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
2.2.2.2	4	65002	15	14	4	0	0	00:09:39	2

7. 检查主机间的连通性

```
VPCS[2]> ping 10.0.0.254
10.0.0.254 icmp_seq=1 ttl=62 time=37.500 ms
10.0.0.254 icmp_seq=2 ttl=62 time=42.500 ms
10.0.0.254 icmp_seq=3 ttl=62 time=37.500 ms
10.0.0.254 icmp_seq=4 ttl=62 time=42.500 ms
10.0.0.254 icmp_seq=5 ttl=62 time=37.500 ms

VPCS[2]> ping 172.16.3.254
172.16.3.254 icmp_seq=1 ttl=62 time=45.001 ms
172.16.3.254 icmp_seq=2 ttl=62 time=35.001 ms
172.16.3.254 icmp_seq=3 ttl=62 time=35.001 ms
172.16.3.254 icmp_seq=4 ttl=62 time=40.001 ms
172.16.3.254 icmp_seq=5 ttl=62 time=42.501 ms
```

四、进阶技能的任务与要求

根据相关的技术与背景知识，并以前面的基本技能训练为基础，请尝试自主完成以下任务：某企业网络及 IP 规划如图 6.9.2 所示，请按以下要求完成路由的规划与配置：

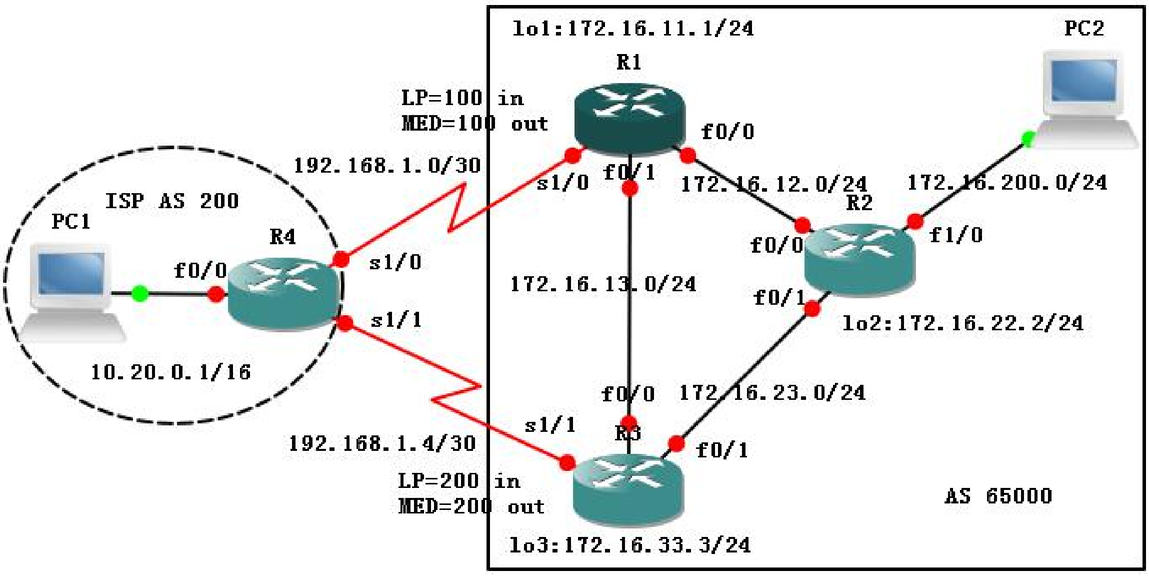

图 2 “BGP 的配置与管理”之进阶技能

1）在整个网络中，采用 BGP 实现与 ISP 的连通。

2）在企业网络的两台路由器上对企业网络进行路由汇总。

3）优先选择总部的边缘路由器 R1 作为互联网流量进入企业的入口，并优先选择分公司的边缘路由器 R3 作为访问互联网流量的出口。

五、进阶技能的规划设计与步骤

1. IP 地址规划，见图；

2. IGP 协议规划：在企业网络内部运行单区域 OSPF 路由协议。其中 R1 参与的接口有 fa0/0, fa0/1, lo1；R2 参与的接口有 fa0/0, fa0/1, fa1/0, lo2；R3 参与的接口有 fa0/0, fa0/1, lo3；

3. 路由汇总规划：在 R1 和 R3 上生成指向 null0 接口的静态路由 172.16.0.0/16；

4. BGP 对等体关系规划：R4 与 R1、R3 形成 EBGP 对等体，企业内三台路由器形成全互联的 IBGP 对等体，均使用自身的环回接口；

5. 路由策略规划：如图所示，通过在 R1、R3 上对与 EBGP 对等体 R4 相关的本地优先级（Local Preference）和 MED 值进行设备来控制数据的出入。具体地

R1→R4: MED=100, R1←R4: LP=100

R3→R4: MED=200, R3←R4: LP=200。

六、进阶技能实验结果与分析

1. 基本连通性配置（路由器名称与接口 IP 地址、动态路由协议、静态路由协议、主机 IP 地址相关配置）

主机 IP 地址：

```
VPCS[1]> ip 10.20.0.254/16 10.20.0.1
Checking for duplicate address...
PC1 : 10.20.0.254 255.255.0.0 gateway 10.20.0.1

VPCS[1]> 2
VPCS[2]> ip 172.16.200.254/24 172.16.200.1
Checking for duplicate address...
PC2 : 172.16.200.254 255.255.255.0 gateway 172.16.200.1
```

R1#sh ip int b

Interface	IP-Address	OK? Method Status	Protocol
FastEthernet0/0	172.16.12.1	YES manual up	up

```
FastEthernet0/1            172.16.13.1     YES manual up                    up
Serial1/0                  192.168.1.2     YES manual up                    up
Serial1/1                  unassigned      YES unset  administratively down down
Serial1/2                  unassigned      YES unset  administratively down down
Serial1/3                  unassigned      YES unset  administratively down down
Loopback1                  172.16.11.1     YES manual up                    up
```

R2#sh ip int b

```
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            172.16.12.2     YES manual up                    up
FastEthernet0/1            172.16.23.2     YES manual up                    up
FastEthernet1/0            172.16.200.1    YES manual up                    up
Loopback2                  172.16.22.2     YES manual up                    up
```

R3#sh ip int b

```
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            172.16.13.3     YES manual up                    up
FastEthernet0/1            172.16.23.3     YES manual up                    up
Serial1/0                  unassigned      YES unset  administratively down down
Serial1/1                  192.168.1.6     YES manual up                    up
Serial1/2                  unassigned      YES unset  administratively down down
Serial1/3                  unassigned      YES unset  administratively down down
Loopback3                  172.16.33.3     YES manual up                    up
```

R4#sh ip int b

```
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            10.20.0.1       YES manual up                    up
FastEthernet0/1            unassigned      YES unset  administratively down down
Serial1/0                  192.168.1.1     YES manual up                    up
Serial1/1                  192.168.1.5     YES manual up                    up
Serial1/2                  unassigned      YES unset  administratively down down
Serial1/3                  unassigned      YES unset  administratively down down
```

R1# sh ip route | exclude C

```
     172.16.0.0/16 is variably subnetted, 8 subnets, 2 masks
O       172.16.200.0/24 [110/21] via 172.16.13.3, 00:06:50, FastEthernet0/1
O       172.16.33.0/24 [110/11] via 172.16.13.3, 00:06:24, FastEthernet0/1
O       172.16.22.0/24 [110/21] via 172.16.13.3, 00:06:40, FastEthernet0/1
O       172.16.23.0/24 [110/20] via 172.16.13.3, 00:06:50, FastEthernet0/1
S       172.16.0.0/16 is directly connected, Null0
```

R2#sh ip route | exclude C

```
     172.16.0.0/24 is subnetted, 7 subnets
O       172.16.33.0 [110/11] via 172.16.23.3, 00:05:34, FastEthernet0/1
O       172.16.13.0 [110/20] via 172.16.23.3, 00:05:49, FastEthernet0/1
O       172.16.11.0 [110/21] via 172.16.23.3, 00:05:49, FastEthernet0/1
```

R3#sh ip route | exclude C

```
     172.16.0.0/16 is variably subnetted, 8 subnets, 2 masks
O       172.16.200.0/24 [110/11] via 172.16.23.2, 00:04:27, FastEthernet0/1
O       172.16.22.0/24 [110/11] via 172.16.23.2, 00:04:27, FastEthernet0/1
O       172.16.12.0/24 [110/20] via 172.16.23.2, 00:04:27, FastEthernet0/1
                       [110/20] via 172.16.13.1, 00:04:27, FastEthernet0/0
O       172.16.11.0/24 [110/11] via 172.16.13.1, 00:04:27, FastEthernet0/0
S       172.16.0.0/16 is directly connected, Null0
     192.168.1.0/30 is subnetted, 1 subnets
```

2. BGP 对等体配置

配置:

R1#sh run | begin router bgp

```
router bgp 65000
 no synchronization
 network 172.16.0.0
 neighbor 172.16.22.2 remote-as 65000
 neighbor 172.16.22.2 update-source Loopback1
 neighbor 172.16.22.2 next-hop-self
 neighbor 172.16.33.3 remote-as 65000
 neighbor 172.16.33.3 update-source Loopback1
 neighbor 172.16.33.3 next-hop-self
 neighbor 192.168.1.1 remote-as 200
 no auto-summary
```

R2#sh run | begin router bgp

```
router bgp 65000
 no synchronization
 neighbor 172.16.11.1 remote-as 65000
 neighbor 172.16.11.1 update-source Loopback2
 neighbor 172.16.11.1 next-hop-self
 neighbor 172.16.33.3 remote-as 65000
 neighbor 172.16.33.3 update-source Loopback2
 neighbor 172.16.33.3 next-hop-self
 no auto-summary
```

R3#sh run | begin router bgp

```
router bgp 65000
 no synchronization
 network 172.16.0.0
 neighbor 172.16.11.1 remote-as 65000
 neighbor 172.16.11.1 update-source Loopback3
 neighbor 172.16.11.1 next-hop-self
 neighbor 172.16.22.2 remote-as 65000
```

neighbor 172.16.22.2 update-source Loopback3

neighbor 172.16.22.2 next-hop-self

neighbor 192.168.1.5 remote-as 200

no auto-summary

R4#sh run | begin router bgp

router bgp 200

no synchronization

network 10.20.0.0 mask 255.255.0.0

neighbor 192.168.1.2 remote-as 65000

neighbor 192.168.1.6 remote-as 65000

no auto-summary

结果:

R1#sh ip route | include B

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

B 10.20.0.0 [20/0] via 192.168.1.1, 00:01:59

R2#sh ip route | include B

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

B 172.16.0.0/16 [200/0] via 172.16.33.3, 00:01:21

B 10.20.0.0 [200/0] via 172.16.33.3, 00:01:49

R3#sh ip route | include B

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

B 10.20.0.0 [20/0] via 192.168.1.5, 00:02:51

R4#sh ip route | include B

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

B 172.16.0.0/16 [20/0] via 192.168.1.6, 00:02:04

```
R1#sh ip bgp
BGP table version is 5, local router ID is 172.16.11.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
* i10.20.0.0/16     172.16.33.3              0    100      0 200 i
*>                  192.168.1.1              0             0 200 i
*> 172.16.0.0       0.0.0.0                  0         32768 i
* i                 172.16.33.3              0    100      0 i
```

```
R2#sh ip bgp
BGP table version is 4, local router ID is 172.16.22.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*>i10.20.0.0/16     172.16.33.3              0    100      0 200 i
* i                 172.16.11.1              0    100      0 200 i
* i172.16.0.0       172.16.11.1              0    100      0 i
*>i                 172.16.33.3              0    100      0 i
```

```
R3#sh ip bgp
BGP table version is 3, local router ID is 172.16.33.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 10.20.0.0/16     192.168.1.5              0             0 200 i
* i                 172.16.11.1              0    100      0 200 i
* i172.16.0.0       172.16.11.1              0    100      0 i
*>                  0.0.0.0                  0         32768 i

R4#sh ip bgp
BGP table version is 3, local router ID is 192.168.1.5
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 10.20.0.0/16     0.0.0.0                  0         32768 i
*  172.16.0.0       192.168.1.2              0             0 65000 i
*>                  192.168.1.6              0             0 65000 i
```

*

\>

检查现有路由情况：

双向 Ping：

```
VPCS[1]> ping 172.16.200.254
172.16.200.254 icmp_seq=1 ttl=61 time=93.600 ms
172.16.200.254 icmp_seq=2 ttl=61 time=124.800 ms
172.16.200.254 icmp_seq=3 ttl=61 time=124.800 ms
172.16.200.254 icmp_seq=4 ttl=61 time=93.600 ms
172.16.200.254 icmp_seq=5 ttl=61 time=109.200 ms

VPCS[1]> 2
VPCS[2]> ping 10.20.0.254
10.20.0.254 icmp_seq=1 ttl=61 time=124.801 ms
10.20.0.254 icmp_seq=2 ttl=61 time=124.801 ms
10.20.0.254 icmp_seq=3 ttl=61 time=124.800 ms
10.20.0.254 icmp_seq=4 ttl=61 time=124.800 ms
10.20.0.254 icmp_seq=5 ttl=61 time=93.600 ms
```

双向 Trace：

```
VPCS[2]> trace 10.20.0.254
trace to 10.20.0.254, 8 hops max, press Ctrl+C to stop
 1   172.16.200.1   31.200 ms  15.600 ms  15.600 ms
 2   172.16.23.3   78.000 ms  62.400 ms  62.400 ms
 3   192.168.1.5   109.200 ms  78.000 ms  93.600 ms
 4   *10.20.0.254   124.800 ms (ICMP type:3, code:3, Destination port unreachabl
e)

VPCS[2]> 1
VPCS[1]> trace 172.16.200.254
trace to 172.16.200.254, 8 hops max, press Ctrl+C to stop
 1   10.20.0.1   31.200 ms  31.200 ms  15.600 ms
 2   192.168.1.6   46.800 ms  46.800 ms  62.400 ms
 3   172.16.23.2   124.800 ms  93.600 ms  93.600 ms
 4   *172.16.200.254   124.800 ms (ICMP type:3, code:3, Destination port unreach
able)
```

PC2→R2→R1→R4→PC1

PC1→R4→R3→R2→PC2

3. 路由策略部署（仅改变 R1 与 R3 的配置，然后在 R4 上重置 BGP 进程）

R1：

router bgp 65000

neighbor 192.168.1.1 route-map LP_100 in

```
neighbor 192.168.1.1 route-map MED_100 out
!
access-list 1 permit 10.20.0.0 0.0.255.255
access-list 2 permit 172.16.0.0 0.0.255.255
!!
route-map MED_100 permit 10
 match ip address 2
 set metric 100
!
route-map MED_100 deny 20
!
route-map LP_100 permit 10
 match ip address 1
 set local-preference 100
!
route-map LP_100 deny 20

R3:
router bgp 65000
 neighbor 192.168.1.5 route-map LP_200 in
 neighbor 192.168.1.5 route-map MED_200 out
 no auto-summary
!
access-list 1 permit 10.20.0.0 0.0.255.255
access-list 2 permit 172.16.0.0 0.0.255.255
!
!
route-map MED_200 permit 10
 match ip address 2
 set metric 200
!
route-map MED_200 deny 20
!
route-map LP_200 permit 10
 match ip address 1
 set local-preference 200
!
route-map LP_200 deny 20

R4#clear ip bgp *
R4#
*Mar  1 01:08:44.047: %BGP-5-ADJCHANGE: neighbor 192.168.1.2 Down User reset
*Mar  1 01:08:44.055: %BGP-5-ADJCHANGE: neighbor 192.168.1.6 Down User reset
```

***Mar 1 01:08:44.295: %BGP-5-ADJCHANGE: neighbor 192.168.1.2 Up**

***Mar 1 01:08:44.703: %BGP-5-ADJCHANGE: neighbor 192.168.1.6 Up**

结果:

```
R1#sh ip bgp
BGP table version is 9, local router ID is 172.16.11.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*>i10.20.0.0/16     172.16.33.3              0    200      0 200 i
*                   192.168.1.1              0    100      0 200 i
*> 172.16.0.0       0.0.0.0                  0         32768 i
* i                 172.16.33.3              0    100      0 i
```

```
R2#sh ip bgp
BGP table version is 7, local router ID is 172.16.22.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*>i10.20.0.0/16     172.16.33.3              0    200      0 200 i
* i172.16.0.0       172.16.11.1              0    100      0 i
*>i                 172.16.33.3              0    100      0 i
```

```
R3#sh ip bgp
BGP table version is 5, local router ID is 172.16.33.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 10.20.0.0/16     192.168.1.5              0    200      0 200 i
* i172.16.0.0       172.16.11.1              0    100      0 i
*>                  0.0.0.0                  0         32768 i
```

```
R4#sh ip bgp
BGP table version is 3, local router ID is 192.168.1.5
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 10.20.0.0/16     0.0.0.0                  0         32768 i
*  172.16.0.0       192.168.1.6            200             0 65000 i
*>                  192.168.1.2            100             0 65000 i
```

BGP 路由发生了改变:

R1#sh ip route | include B

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

B 10.20.0.0 [200/0] via 172.16.33.3, 00:10:23

R2#sh ip route | include B

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

B 172.16.0.0/16 [200/0] via 172.16.33.3, 00:46:05

B 10.20.0.0 [200/0] via 172.16.33.3, 00:13:31

R3#sh ip route | include B

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

B 10.20.0.0 [20/0] via 192.168.1.5, 00:13:56

R4# sh ip route | include B

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

B 172.16.0.0/16 [20/100] via 192.168.1.2, 00:12:03

```
VPCS[2]> trace 10.20.0.254
trace to 10.20.0.254, 8 hops max, press Ctrl+C to stop
 1   172.16.200.1   31.200 ms  31.200 ms  31.200 ms
 2   172.16.23.3   93.600 ms  46.800 ms  46.800 ms
 3   192.168.1.5   93.600 ms  109.200 ms  93.600 ms
 4   *10.20.0.254   140.400 ms (ICMP type:3, code:3, Destination port unreachabl
e)

VPCS[2]> 1
VPCS[1]> trace 172.16.200.254
trace to 172.16.200.254, 8 hops max, press Ctrl+C to stop
 1   10.20.0.1   31.200 ms  31.200 ms  31.200 ms
 2   192.168.1.2   109.201 ms  78.000 ms  62.400 ms
 3   172.16.13.3   93.600 ms  78.000 ms  78.000 ms
 4   172.16.23.2   140.401 ms  109.200 ms  124.800 ms
 5   *172.16.200.254   140.400 ms (ICMP type:3, code:3, Destination port unreach
able)
```

PC2→R2→R3→R4→PC1

PC1→R4→R1→R3→R2→PC2

七：思考题：

1. 在声明 EBGP 对等体时，是否应该定义 update-source 选项？为什么？
2. BGP 路由器在使用 network 命令宣告网络时，是否能宣告一个并不是自己直连的网络？为什么？
3. 是不是自治系统中所有的路由器都应该运行 BGP？为什么？

八、教师评语：

实验成绩：　　　　　　　　　　教师：（签名要全称）　　　年　　月　　日

注：1。此模板为专业实验报告的基本要求，若有特殊要求的实验，可在此模板基础上增加，但不可减少。

2．实验报告必须在学生提交报告后一星期内批改。

另附创新技能方案及结果分析

说明：

① 上下页边距改成 2 厘米，左边距为 2.0 厘米，右边距为 1.5 厘米。

② 表格位置为居中

创新活动：

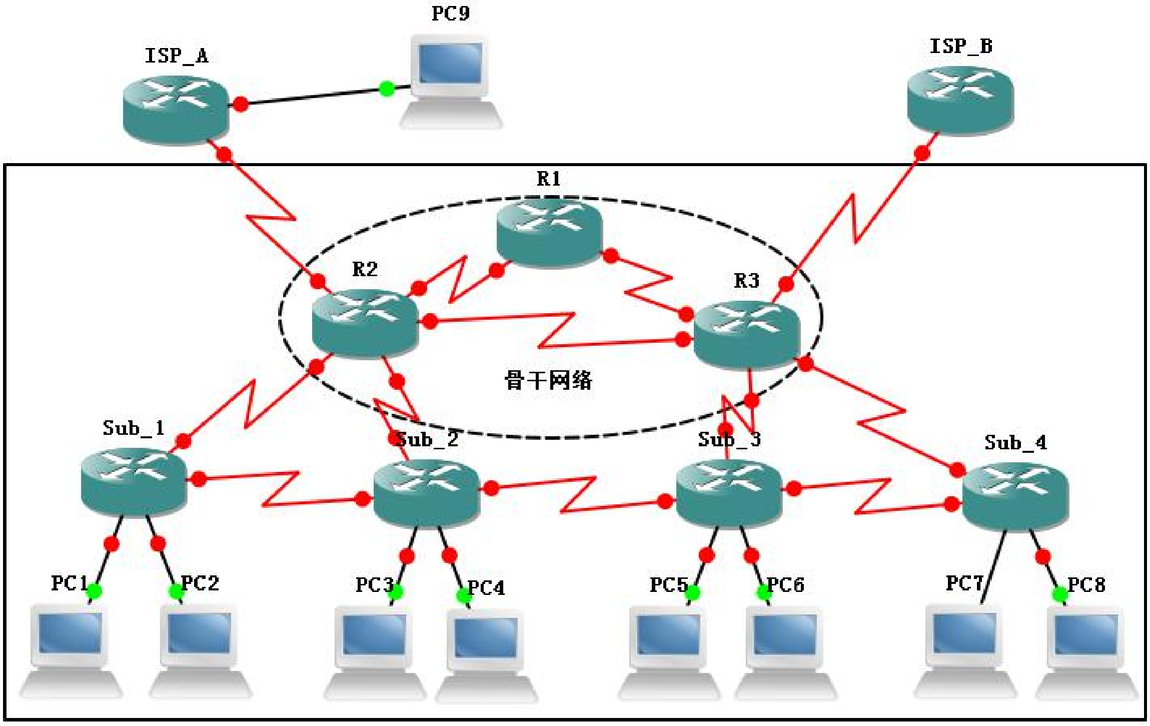

图 3 “BGP 的配置与管理”之创新活动

某企业的网络拓扑如图 6.9.3 所示，企业有较多的分部，以后分部数目还会继续不断的增加，且每个分部的子网规模均超过 200 以上。其中总部的骨干网络有多个到 ISP 的连接，这些连接个数还将不断增加。现有人向你建议了如下的路由规划：

1) 在企业内部, 采用多区域 OSPF 实现动态路由, 其中总公司以及总公司与分公司的连接属于主干区域。
2) 在企业与各 ISP 之间，通过 BGP 路由协议实现自治系统间互联。为达到可以完全操纵路径选择的目的，企业与 ISP 交换完整的路由信息。
3) 在企业网络内部，所有路由器都运行 BGP 路由协议，并与任意其它路由器形成 IBGP 对等体关系。

请你完成以下工作：

1) 根据该建议，给出路由的具体配置方案并付诸实施；
2) 分析与判断该方案的可行性，分析方案对企业内部非骨干网络的路由器造成的影响
3) 搜集并讨论可能的解决方案。